

Keshav Rajput

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EDUCATION

New York University, Tandon School of Engineering

MS, Computer Science, May 2026. GPA 3.67/4.0

IIIT Delhi

B.Tech, Computer Science and Applied Mathematics; minor in Economics, 2020 - 2024. CGPA 8.19/10

Coursework: Machine Learning, Neuroinformatics, Foundations of Data Science, Advanced Machine Learning, Convex Optimization, Reinforcement Learning

PROJECTS

LFP2Vec audit | Python, PyTorch, wav2vec2, GitHub Actions github.com/happyc0der/lfp2vec-audit, 2026

- Reproduced LFP2Vec (NeurIPS 2025) on public IBL and Allen Neuropixels data: 0.72 balanced accuracy vs 0.68 published.
- Found cross-lab transfer falls to chance (0.24) because of a high-frequency preprocessing mismatch; a 100 Hz low-pass brings it back to 0.46.
- Added calibration under lab shift (ECE 0.56 to 0.35), 3-seed error bars and 300+ tests.

Biohub cell tracking | Python, PyTorch, 3D CNN, Kaggle API 2026

- Kaggle Biohub Cell Tracking: public leaderboard 0.952, about top 1.2% of 3,594 teams.
- Offline harness that splits GPU inference from CPU post-processing, so tuning runs locally against the official scorer.
- 3D-CNN cell-division classifier trained on 134 examples; holdout predicted +0.0047, the leaderboard gave +0.005.

Blockwave | Python, numpy, PyTorch, PPO github.com/happyc0der/blockwave, 2026

- Tetris-style game in pure Python and numpy with no asset files; graphics and audio are generated in code.
- PPO agent trained on an empowerment reward, never shown score or lines; board-state and pixel-only versions.
- 465 tests; results reported with seed variance after two identical 150M-step runs differed by 42%.

Kaggle S6E9, EV purchases | Python, LightGBM, XGBoost, github.com/happyc0der/kaggle-s6e9-ev-purchases, 2026
CatBoost, PyTorch, Optuna

- Kaggle Playground S6E9 (EV purchases): out-of-fold AUC 0.9465, public leaderboard 0.94649 against a leader at 0.94674.
- Nested target encoding feeding LightGBM, XGBoost, CatBoost and an MLP; 669k training rows.
- 190+ logged experiments compared with paired DeLong tests; showed the remaining gap is label noise.

COMPETITIONS

- **Biohub Cell Tracking, Kaggle**, 2026. Public leaderboard 0.952, about top 1.2% of 3,594 teams.
- **Kaggle Playground S6E9**, 2026. AUC 0.9465, 0.00025 behind the leader.
- **Amazon Build, Ship, Shape 2026 (Alexa+ track)**, 2026. Shipped Custodian, an MCP recall-monitoring add-on.

SKILLS

Languages: Python, C, C++, Kotlin, TypeScript, JavaScript, SQL, Java, R, Bash

ML: PyTorch, scikit-learn, LightGBM, XGBoost, CatBoost, Optuna, NumPy, pandas, polars

Data: PySpark, Databricks, SQL, PostgreSQL, SQLite, pandas, Streamlit, Altair, matplotlib

Tools: Git, GitHub Actions, Docker, AWS (CDK, App Runner, DynamoDB), Linux, LaTeX